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ORIGINAL RESEARCH

Exploring the Capacity of Large Language Models to Simulate Students' Scientific Thinking: Insights for Responsive Teaching

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Abstract
The success of responsive science teaching to elicit, foreground, and pursue the substance of students' ideas depends on how teachers can recognize and respond to these ideas and connect them to disciplinary perspectives. One approach to supporting responsive teaching is to design instruction that facilitates students' scientific sensemaking and invites teachers to consider possible student ideas. Generating students' ideas, however, is labor-intensive as it requires understanding of learning content and instructional contexts. In this work, we explore the capacity of several large language models (LLMs), including OpenAI's GPT-5-mini, GPT-4o, Claude's Sonnet 4, Gemini 2.5 Flash, Mistral 7B, and Llama-4-17B, to simulate 8820 students' ideas in science learning activities across domains (life sciences, physics, and chemistry) and grade levels (elementary, middle, and high school) to support instruction planning. Findings indicate that the LLM-simulated responses are realistic and are mostly at the target grade levels for knowledge scope and readability. We observe variations in performance across LLMs and grade levels, with LLMs producing more within-scope ideas for high school lessons and overly complex ideas for lower grades. Interviews with six teachers using the LLM-generated ideas with their own lesson plans reveal that the simulated student responses are overall realistic and align with the lessons' objectives. Responses are most useful when they spark teachers' sensemaking about what students know and suggest relevant instructional scaffolds. We discuss how to improve LLMs' simulations of student thinking and promote responsive science teaching with AI-generated insights.

Keywords Responsive science teaching · Large language models · Lesson planning · Science education

Introduction

Science instruction in K-12 education has historically focused on normative reasoning, academic terminologies, and memorization in place of sensemaking (National Academies of Sciences, 2021; National Research Council, 2012). Teachers can benefit from professional development to attend to the complexity of student ideas and build on student thinking as resources for emergent sensemaking (Davis & Bautista, 2025; Rosebery et al., 2016). These practices reflect *responsive teaching* approaches that foreground and pursue the substance of students' ideas (Hammer et al., 2012; Robertson et al., 2015). The success of responsive teaching depends on how teachers can recognize and respond to students' ideas, connect these ideas to disciplinary perspectives, and prompt students to build on others' ideas (Penuel et al., 2022; Pimentel & McNeill, 2013; Thompson et al., 2016). However, teachers may not always anticipate or notice students' thinking during lesson preparation and enactment (Hutchison & Hammer, 2010; Kang & Anderson, 2015). They may face challenges in synthesizing student perspectives and interests and mobilizing students' everyday experiences as resources for instruction (Lemke, 1990; Silseth & Erstad, 2022; Wells & Arauz, 2006).

One approach to supporting responsive teaching for in-service and pre-service teachers is to invite teachers to design instruction that facilitates students' evolving sensemaking of scientific phenomena and consider possible student ideas in lesson planning and how they might respond to them (Davis & Bautista, 2025; Edelson et al., 2021).

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This approach strengthens teacher sensemaking, defined as the active process of noticing, interpreting, and creating meanings from student ideas and developing understanding of responsive teaching in specific instructional contexts (Robertson & Richards, 2017; Spillane et al., 2002). Generating students' ideas, however, is labor-intensive and challenging when preparing new instruction for teachers. It requires understanding of learning content, instructional contexts, and potential students' responses based on their existing knowledge. In this work, we explore the capacity of several large language models (LLMs), including commercial (OpenAI's GPT-5-mini, GPT-4o, Claude's Sonnet 4, Gemini 2.5 Flash) and open-source LLMs (Mistral 7B, Llama-4-17B), to simulate students' open-ended responses to science learning activities across a variety of domains and grade levels. LLMs have shown promise for simulating student cognitive processes and classroom interactions and offering suggestions to improve instruction (Hu et al., 2025; Lu & Wang, 2024). We investigate whether LLM-simulated responses can serve as resources for teachers to anticipate student ideas and suggest teachers' responses to those ideas. To achieve this, LLM-simulated ideas should capture possible student ideas and align with instructional contexts. We thus build on prior research that has examined the similarity between LLM-generated and student ideas (Lu & Wang, 2024), language authenticity (Milička et al. 2024; Zheng et al. 2025a, b), and students' knowledge (Jin et al., 2025). We investigate the following research questions (RQs):

RQ1 To what extent can LLMs simulate students' responses to open-ended questions in science investigations? Specifically, to what extent do the simulated responses overlap with lesson ideas (RQ1a), demonstrate grade-level language (RQ1b), and show an appropriate knowledge scope (RQ1c)?

RQ2 How do teachers perceive the utility of LLM-simulated ideas for lesson planning and facilitation?

This study has two main contributions to science education and technology research. First, we provide evidence of how LLMs can approximate open-ended student thinking in ways that overlap with teacher and student-generated lesson ideas and largely show appropriate knowledge scopes and language levels. Findings about performance variations across LLMs and grade levels demonstrate the need for comprehensive evaluation before deployment and further effort to fine-tune LLMs to simulate novice reasoning. Second, we demonstrate how LLM simulations can support responsive teaching practices, through externalizing students' ideas, prompting teachers' sensemaking, and suggesting instructional scaffolds. These approaches align with recent calls for AI-enhanced instruction (Seo et al., 2025) where teachers partner with AI to augment and extend instructional support. Our findings inform future analytic tools to prepare pre-service and in-service teachers for responsive science instruction.

Literature Review

Responsive Teaching in Science Classrooms

Responsive teaching entails attending to the substance of students' ideas, connecting them to disciplinary ideas, and pursuing students' thinking (Coffey et al., 2011; Hammer, 1997; Hammer et al., 2012; Robertson et al., 2015). Our conceptualization of responsive teaching builds on both cognitive and sociocultural perspectives. From a cognitive perspective, responsive teaching recognizes students' epistemological resources, defined as fine-grained knowledge elements, that teachers can recognize to assess and support students' learning (Elby & Hammer, 2010). From a sociocultural perspective, learning is a process of meaning-making through social interactions in particular systems and practices (Greeno, 1998). Teachers facilitate and engage in dialogues with students to shape meaning making in science learning (Krist, 2024; Mercer et al., 2019).

A core assumption of responsive teaching is that students bring abundant resources for science sensemaking (Danish & Enyedy, 2007; Louca et al., 2004). These include concrete experiences and intuitions (DiSessa, 1993; Smith et al., 1994) and epistemological resources like argumentation, explanation, and modeling (Elby & Hammer, 2010; Louca et al., 2004; Reiner & Gilbert, 2000). These resources can be fragmented and reflect students' everyday thinking (Hammer & Elby, 2003). When teachers validate and use students' resources to drive investigations, students can develop epistemic agency to shape classroom knowledge building (Stroupe, 2014). Researchers have thus called for instructional design and facilitation to draw on students' initial, intuitive ideas and encourage students to build, extend, and revise these ideas as they make sense of scientific phenomena (Berland et al., 2016, 2020; Penuel et al., 2022; Windschitl et al., 2012). Teachers engage in responsive teaching in interaction with individual learners, small groups, and the whole class (Felton et al., 2022; Dyer & Sherin, 2016). They can elicit student thinking (Harris et al., 2012; Robertson et al., 2015), prompt students to elaborate on ideas or explain phenomena they might have missed (Manz & Suárez, 2018; Watkins et al., 2020), help students wrestle with uncertainties about which questions to pursue (Chen, 2022; Y.-C. Chen & Qiao, 2020), and invite students to critique and build on others' ideas (Michaels et al., 2008). These moves differ from less responsive practices that merely evaluate correctness, steer students toward predetermined response, or elicit comments without substantive follow-up (Grinath & Southerland, 2019; Thompson et al., 2016).

Responsive teaching is inherently complex. To enact responsive teaching, teachers need to practice sensemaking to construct meanings about specific aspects of student thinking and teaching experiences, negotiate key questions about what it means to build on students' noncanonical ideas, and situate such noticing within their instructional contexts (Robertson & Richards, 2017). Teachers must relate students' thinking to the structure of the learning tasks and develop contexts to test preliminary interpretations of student thinking (Dyer & Sherin, 2016). They need to facilitate and consolidate different student perspectives in ways that are meaningful to the classroom communities (Silseth & Erstad, 2022; Wells & Arauz, 2006). Furthermore, epistemic uncertainty—students' awareness of their inability to make sense of a phenomenon or form a scientific argument—can serve as resources for learning (Chen, 2022). However, teachers might avoid uncertainty (Kampourakis, 2018) or treat it superficially as a source to induce curiosity rather than pushing student thinking (Starrett et al., 2024).

To promote responsive teaching, teachers can contextualize science ideas and invite students to conduct scientific inquiries and come up with ideas and solutions relevant to their experiences (Miller & Krajcik, 2019; Sevian et al., 2018). For example, the storyline approach to science curriculum development (Penuel et al., 2022; Reiser et al., 2021) involves students and teachers as collaborators in knowledge building and makes scientific meaning-making coherent from students' perspectives (Penuel & Reiser, 2018). Central to this approach is the anchoring of instruction in questions connected to students' experiences (Kubiatko et al., 2021; Sevian et al., 2018; Swirski et al., 2018); linking students' initial ideas and practices with science phenomena (Guy-Gaytán et al., 2019); and promotion of students' agency and participation (Michaels et al., 2008; Reiser et al., 2021; Thompson et al., 2016; Windschitl & Calabrese Barton, 2016). Additionally, instructional materials can include examples of partially complete or incorrect student ideas and how to build instruction dynamically on these ideas (Berland et al., 2020; Penuel et al., 2022). These resources can support teachers' sensemaking, by helping teachers anticipate possible student responses and prepare more targeted instructional moves to leverage students' thinking.

However, teachers face challenges in developing these examples at scale, as they need to conduct careful analysis of classroom interactions and learning objectives. This challenge motivates us to explore the possibility of leveraging LLMs to simulate students' ideas and support responsive teaching facilitation.

Using LLMs to Simulate Students' Ideas

Simulations of student thinking provide opportunities for teachers to practice instructional dialogues, invite students to collaborate with a simulated learning companion, and allow developers to test instruction (VanLehn et al., 1994). Simulation approaches involve representing students' learning processes and cognitive states to predict their future behaviors and performance (Chrysafiadi & Virvou, 2013; Li et al., 2011). These approaches include cognitive (e.g., knowledge, skills, decision-making), affective (e.g., motivation, emotion), metacognitive (e.g., self-regulation, self-explanation, self-assessment), and communication factors (Chrysafiadi & Virvou, 2013). Researchers have explored rule-based methods, where instructional designers can specify rules to model student thinking (Anderson & Schunn, 2013; MacLellan et al., 2015), or machine learning approaches that simulate student performance from historical responses (Abdelrahman et al., 2023; Käser et al., 2017). However, these approaches require human development, validation, and large training datasets, and are thus labor-intensive to implement.

LLMs offer new opportunities to address this challenge. LLMs that possess flexible content generation and conversational interaction capabilities have shown promise in education settings to provide personalized feedback to students (Cao et al., 2025; Zhao et al., 2025), serve as intelligent tutors or instructors (Limo et al., 2023; Hao et al., 2026), and assist educators in developing lesson plans (Şimşek, 2025), learning tasks (Küchemann et al., 2023), and assessment items (Chan et al., 2025). Researchers have also used LLMs to simulate learning behaviors (Binz et al. 2025; Jin et al. 2025; Zheng et al. 2025a, b). Efforts that leverage LLMs to create student simulations have primarily relied on cognitive modeling that incorporates academic-related information, such as grade levels, prior knowledge, and learning histories, to simulate students' system interactions and responses to test items (Xu et al. 2025; Zheng et al. 2025a, b). Beyond cognitive factors, researchers have incorporated motivation, personality traits, and affective states to create more holistic student simulations (Bhowmik et al., 2024; Pan et al., 2025). For example, Jin et al. (2025) designed student agents with different knowledge-level specifications and motivation, self-efficacy, stress levels, and goal commitment profiles. The agents served dual purposes: providing instructional rehearsal opportunities and enhancing pedagogical skills for teachers.

To evaluate simulation performance, researchers have employed computational approaches with focus on cognitive aspects (Xu et al. 2025; Zheng et al. 2025a, b). For example, scholars have assessed the correlation between LLM-generated and students' responses (Lu & Wang, 2024); alignment in knowledge scope between the simulated responses and students' cognitive profiles (Jin et al., 2025; Xu et al., 2025); and LLMs' language to reflect age-appropriate development (Milička et al., 2024). Notably, prior evaluations of student simulations have primarily examined interactions within intelligent tutoring systems (Xu et al., 2025; Liu et al., 2024), with few examples of simulating student thinking and responses in authentic classroom interactions. Additionally, researchers have rarely involved teachers in providing feedback.

In this study, we explore the potential of LLMs to simulate students' responses in classroom-based science instruction. This use case offers promise for enhancing responsive teaching, as opportunities to anticipate and make sense of simulated student ideas can support teachers to elicit, understand, and better respond to these ideas in real time. We further investigate whether and how teachers can derive pedagogical insights from the simulated ideas.

Methods

Overall Procedures & Data Sources

RQ1: LLMs Performance

Two research phases mapped onto the RQs (Fig. 1). In the first phase, to answer RQ1, we evaluated the ability of different LLMs to simulate student responses in science learning activities. We included state-of-the-art, commercial LLMs (GPT-5-mini, GPT-4o, Claude Sonnet 4, Gemini 2.5 Flash) and smaller, open-source alternatives (Mistral 7B, Llama 4-17B). These models have demonstrated good performance in education use cases (e.g., Huang et al., 2024; Varasteh Nezhad et al., 2024). They allowed us to explore both proprietary systems with extensive pretraining and less resource-intensive models that might be more practical for classroom use. We examined (a) the *semantic similarity* between LLM-generated student ideas and ideas sourced from teachers and students in classroom implementation (hereafter referred to as lesson ideas), (b) whether the generated ideas demonstrated the appropriate *language level*, and (c) whether they were within grade-level *knowledge scope*. These investigations build on evaluations of LLM simulations in prior work (e.g., Jin et al. 2025; Xu et al. 2025; Zheng et al. 2025a, b).

For this, we created a dataset consisting of suggested student ideas in OpenSciEd lesson plans (OpenSciEd, 2022). OpenSciEd lessons are designed to align with the Next Generation Science Standards (NGSS), are widely adopted (e.g., Andersen et al., 2024; Edelson et al., 2021; Lowell et al., 2024) and provide examples of student thinking that are infrequently featured in other lesson plans. The suggested student ideas in the OpenSciEd lesson plans were created through pilot data collection and consultation with teachers about how students might respond to the learning activities (Edelson et al., 2021). We selected nine learning units across three domains (life sciences, physics, and chemistry) and three grade levels (elementary, middle, and high school) and extracted 49 lesson plans and the ideas from these lesson plans to create a dataset of lesson ideas (Appendix A outlines the lesson plans). Each lesson plan consisted of 28 ideas on average, $SD = 10$. We instructed several LLMs

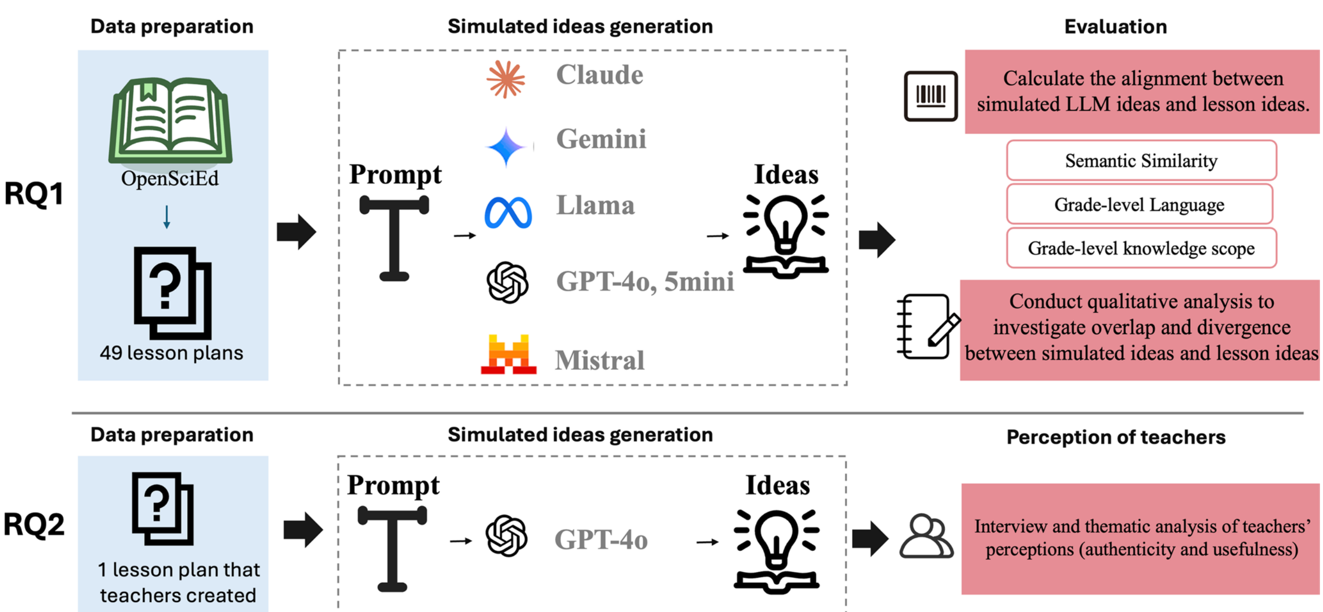


Fig. 1 Overall research procedures

You are simulating responses from {row['grade']} students during a {row['domain']} lesson.

Background
Learning Standard: {row['learning_standards']}
Prior Instruction: {row['prior_instruction']}
Activity Description: {row['activity_description']}
Task for Students: {row['task']}

Instruction
Generate 30 different student responses that represent the rich diversity of student thinking. Each response addresses one of the questions and draws from students' varied resources and experiences.

Student responses can:
1.Combine everyday thinking (leveraging personal, environmental, and social resources) with scientific terminology, use purely everyday thinking, or use primarily scientific terminology
2.Incorporate ideas from class activities or build on ideas that originate from other students
3.Be correct, partially correct, or incorrect
4.Show confusion when understanding is developing
5.Reflect prior conceptions (including misconceptions) that serve as valuable resources for learning

Include a mix of:
1.Correct ideas well-aligned with the standard
2.Partially correct ideas showing developing understanding
3.Incorrect ideas or prior conceptions that students might realistically bring to the topic
Use realistic student language (short, informal, sometimes fragmented) and vary reasoning styles from casual observations to more systematic scientific thinking.

Output Format
Return the student ideas in a single line, separated by semicolons (;) only. Do not include numbering, bullets, or extra text. Example: idea1; idea2; idea3

Fig. 2 Prompt for idea generation

to generate 30 simulated ideas per lesson (1470 ideas per LLM; total 8820 ideas for six LLMs).

We generated the simulated ideas based on detailed descriptions of the learning activities from the lesson plans. The prompt (Fig. 2) to the LLMs followed the structure: *Learning Standards* (the science learning standards that the lesson covered and learning goals), *Prior Instruction* (what students did in the previous lesson), *Activity Description* (description of the learning activities and teachers' guiding questions), *Task Description* (specifying the task, the grade level, and learning domain), and *Simulation Instruction* (instruction to simulate the language and reasoning styles, including possible partial or incorrect scientific reasoning). The instruction to include everyday and scientific thinking reflects responsive teaching frameworks that recognize students' multiple epistemological resources (Elby & Hammer, 2010; Robertson et al., 2015). Appendix B shows example outputs.

RQ2: Teachers' Perceptions of the Simulated Ideas

In the second phase (RQ2), we interviewed six teachers in audio-recorded, semi-structured interviews (45 min each). We recruited from our network of science teachers and selected those with a range of teaching experiences, grade levels, and subjects (Table 1). We followed Malterud et al.'s (2016) principle of information power to justify the sample size, given the research's narrow aim and relevance of participants' experience. We collected teachers' perceptions of the LLM-generated ideas based on their own lesson plans and how they might use the ideas to support lesson planning and instruction.

Table 1 Teacher demographics

Name (pseudonym)	Gender	Grade level	Lesson plan	Years teaching	State they teach in
Rylie	Female	Middle	Physics	3	Utah
Allan	Male	Middle	Physics	6	Utah
John	Male	Middle	Physics	2	Utah
Sam	Female	High school	Environmental science	15	California
Alyssa	Female	Elementary	Life science	2	California
Jan	Female	High school	Environmental science	1	California

During the interviews, we asked teachers to bring a lesson plan that they were preparing. Using teachers' own lesson plans instead of OpenSciEd lessons ensured that the simulated ideas were familiar and potentially useful for teachers. We used an LLM (GPT-4o, which showed balanced performance in the first phase of LLM evaluation) in two tasks: (1) generating student ideas in response to the lesson plan activities, and (2) suggesting facilitation prompts, aimed at improving teachers' responsiveness to student thinking given the generated ideas (Fig. 3; see prompts in Appendix C). We asked teachers to verbalize their thoughts on the LLM-generated ideas and suggestions, such as whether the ideas aligned with the learning goals and reflected what their students might say. Teachers provided feedback on whether and how the ideas and facilitation prompts were helpful for instructional planning. The procedures received Institutional Review Board approval.

Analyses

Evaluating LLM-generated Student Ideas (RQ1)

For each LLM, we implemented a five-step evaluation pipeline: creating idea clusters, calculating semantic similarity, evaluating language appropriateness, assessing knowledge scope, and qualitatively analyzing the divergence from lesson ideas.

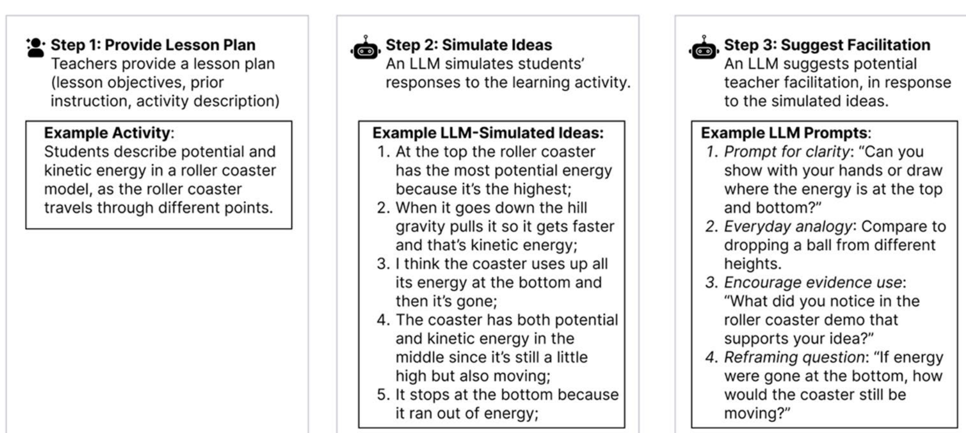
First, we created clusters of LLM-generated ideas. We converted each idea into a vector representation using a sentence embedding model (OpenAI's text-embedding-3-large). We then grouped the vectors using agglomerative clustering with average linkage and cosine distance (distance threshold = 0.30; equivalent to cosine similarity >= 0.70; implemented in scikit-learn). We used clustering to enable more efficient comparisons between lesson ideas and LLMs' idea clusters. This is because LLMs often generated similar ideas to the same questions (e.g., "Fires burn underground in peat", "Peat smolder underground"). Comparing lesson ideas and cluster centroids would be more computationally efficient than comparisons with all generated ideas.

Second, for *semantic similarity* (RQ1a), we compared the clusters of LLM-generated ideas with the lesson ideas. Each cluster was represented by a centroid, which was the average of the embedding vectors of all ideas within that cluster. For each lesson idea, we calculated its cosine similarity with all cluster centroids and retained the highest value as its best match. Cosine similarity measures the similarity between two vectors, by calculating the cosine of the angle between them. Its values ranged between −1 and 1, with higher values indicating higher degrees of text similarity.

Third, we calculated whether the LLM-generated ideas were consistent with the prompt in terms of *language level* (RQ1b). We calculated the Flesch–Kincaid Grade Level for each set of LLM-generated student ideas within a given unit. If the grade level was less than or equal to the target students' actual grade, the idea was labeled as within the language level; otherwise, it was labeled as out of the language level.

Fourth, for *knowledge scope* (RQ1c), we adopted an LLM-as-a-judge approach (Zheng et al., 2023) and prompted an LLM (GPT-5) to determine whether the generated student ideas exceeded the expected grade-level boundaries (Appendix D). If the concepts and reasoning expressed in the ideas aligned with the grade-band expectations specified by the NGSS or corresponding lesson objectives, the idea was coded as within the knowledge scope; otherwise, it was coded as out of scope. A researcher with a background in science education coded 20% of the data to determine knowledge scope alignment and achieved substantial inter-rater agreement with the LLM's classification (Cohen's κ = 0.69). We acknowledged that there could be systematic biases when an LLM was judging output from other LLMs trained on similar data (e.g., GPT-4o, GPT-5-mini). For robustness check, we used Claude-Sonnet-4 to evaluate the knowledge scope of the same data subset. This LMM also showed substantial inter-rater agreement with the same researcher, Cohen's κ = 0.64. We used GPT-5 for the analysis.

Fig. 3 Illustrative teacher interview tasks



To evaluate the LLMs' performance, we fitted linear mixed models to compare the semantic similarity (outcome variable) across LLMs, domains, and grade levels (fixed effects in separate LMMs) using the R package lme4 (Bates et al., 2015). This approach accounted for non-independent observations, since each learning activity appeared multiple times in the dataset due to prompting for the six LLMs. We included activity ID as random intercept to model within-activity correlations. For the binary variables (language level, knowledge scope), we fitted generalized linear mixed models with a binomial distribution and logit link and included activity ID as random intercept. We used likelihood ratio tests to determine if models with the fixed effects provided a better fit for the data than those without them. We estimated post-hoc Tukey pairwise comparisons using the emmeans package (Lenth, 2023). We reported odds ratio (OR) and 95% confidence interval (CI) as a measure of effect size.

In addition, for each LLM, we evaluated performance across domains with Kruskal-Wallis and Dunn's test (in case the data violated normality and homogeneity of variance assumptions) and ANOVA tests with post-hoc Tukey's HSD test otherwise. We conducted Chi-square tests to compare the proportions within language level and knowledge scope by grade levels and domains. To address Type I errors in post-hoc comparisons, we used the Holm-Bonferroni correction. For some within-model analyses, at least one grade or domain level showed complete separation (e.g., 100% of responses were within knowledge scope), which led to zero expected counts in the omnibus chi-square test. In those cases, we used pairwise Fisher's exact tests.

Finally, we qualitatively compared LLM-generated and lesson ideas to identify areas of divergence. This involved feeding both idea corpora to an LLM (GPT-5) to "list 3–5 bullet points for overlap and divergence between the ideas for each lesson". Two researchers then reviewed all LLM's analyses against the input (lesson and simulated ideas) to correct errors, including a few instances where GPT-5 generated details that were not present in the input. The researchers collaboratively generated keywords and themes of divergence patterns. This involved two rounds of iterative coding, where researchers independently reviewed the data and the LLM summaries, individually proposed themes with examples from the data, and discussed interpretations to reach consensus.

Analyzing Teachers' Perceptions (RQ2)

We analyzed interview data using thematic analysis (Braun & Clarke, 2012) to identify teachers' perceptions, including perceived usefulness of the generated ideas and suggestions for integrating LLM simulations into lesson planning. Two researchers conducted open coding of all transcripts to generate initial keywords based on their reflectiveness of participants' experiences, richness, repetition, rationale, repartee (insightfulness), and regal (importance; Naeem et al., 2023) and codes. We refined the codes and organized them into themes through moderation in two meetings and asynchronous discussion. The researchers double-coded the remaining transcripts and resolved discrepancies through discussion. To establish credibility, we conducted member checking. We sent the theme summary to teacher interviewees and requested their feedback and clarification. This allowed us to provide more lesson contexts for the write-up.

Results

RQ1: Evaluating LLM-Generated Student Ideas

RQ1a: Semantic Similarity

Across 49 lesson plans spanning science domains and grade levels, the mean cosine similarity reflecting the overlap between LLM-generated and lesson ideas was 0.52 ($SD=0.07$). A likelihood ratio test indicated that the model including LLM (as fixed effect) provided a significantly better fit for the data than a model without the effect, with larger LLMs achieving slightly higher values than smaller models ($\chi^2(5)=218.69$, $p<.001$; Fig. 4). Specifically, all LLMs performed better than Mistral-7B (reference category; $M=0.47$, $SD=0.06$): GPT-5 mini ($M=0.54$, $SD=0.06$, $\beta=0.06$, $SE=0.01$, $t=14.24$, $p<.001$), GPT-4o ($M=0.54$, $SD=0.06$, $\beta=0.06$, $SE=0.01$, $t=13.97$, $p<.001$), Gemini-2.5-Flash ($M=0.55$, $SD=0.06$, $\beta=0.07$, $SE=0.01$, $t=16.00$, $p<.001$), Claude-Sonnet-4 ($M=0.54$, $SD=0.06$, $\beta=0.06$, $SE=0.01$, $t=14.00$, $p<.001$) and Llama-4 ($M=0.52$, $SD=0.07$, $\beta=0.05$, $SE=0.01$, $t=10.71$, $p<.001$).

Domain did not significantly predict semantic similarity based on results from a likelihood ratio test ($\chi^2(2)=3.83$, $p=.15$). Descriptively, LLMs were able to simulate more similar ideas in physics than in life sciences and chemistry (physics: $M=0.55$, $SD=0.08$; life sciences: $M=0.51$, $SD=0.06$; chemistry: $M=0.52$, $SD=0.06$). Within each LLM, Kruskal–Wallis tests showed no statistically significant differences across domains ($p>.05$; Fig. 5).

Finally, cosine similarity differed significantly by grade level (elementary: $M=0.54$, $SD=0.07$; middle school: $M=0.55$, $SD=0.07$; high school: $M=0.50$, $SD=0.05$; $\chi^2(2)=10.94$, $p=.004$; Fig. 6). Both elementary ($\beta=0.05$, $SE=0.02$, $t=2.75$, $p=.01$) and middle school levels ($\beta=0.05$, $SE=0.02$, $t=2.92$, $p=.01$) had higher similarity than high school (reference category). ANOVA tests further revealed significant grade-level differences within each LLM:

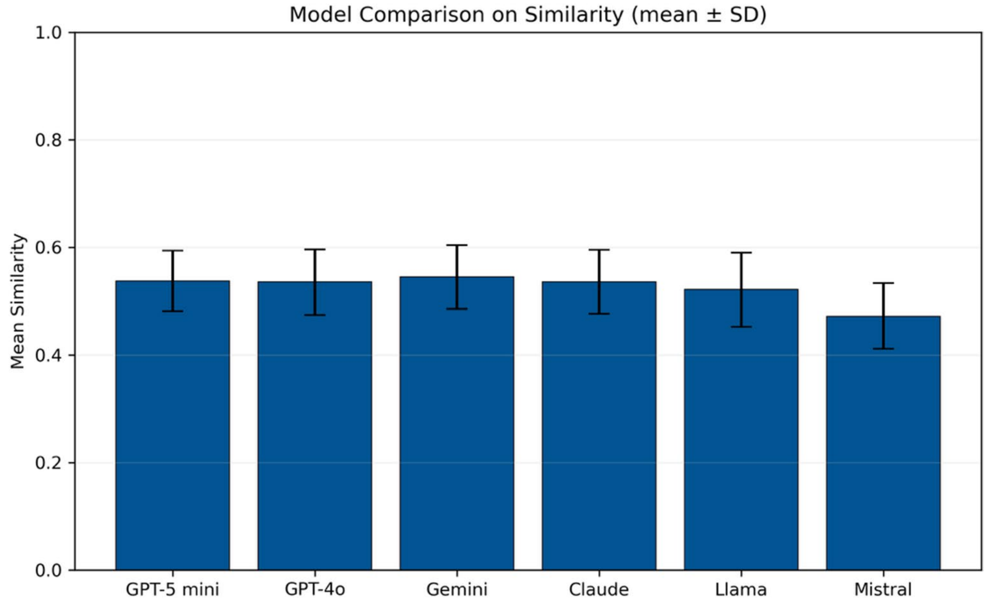


Fig. 4 Semantic similarity between simulated and lesson ideas across LLMs

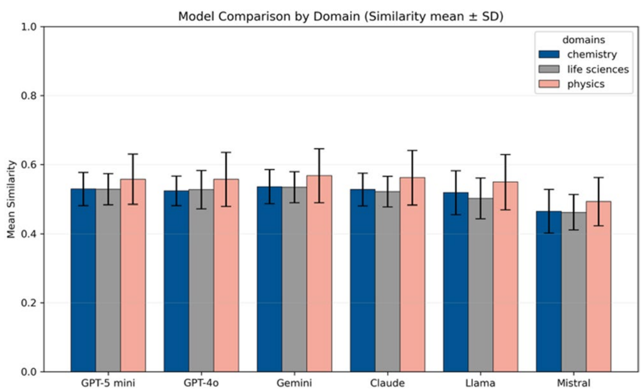


Fig. 5 Semantic similarity between simulated and lesson ideas across six models by domain

GPT-5-mini (F=5.19, p =.01), GPT-4o (F=5.55, p =.01), Gemini-2.5-Flash (F=5.78, p =.01), Llama-4 (F=4.46, p =.02), Claude-Sonnet-4 (F=4.39, p =.02), and Mistral-7B (F=5.79, p=.01).

These results indicated that LLMs might show weaker performance in matching the lessons' learning objectives for higher grade levels. For example, in an elementary chemistry lesson on water quality, when asked, "How do you think we could make the unhealthy water become healthy again?", an idea generated by Gemini-2.5-Flash was: "Maybe the unhealthy water has too much dirt in it, we could scoop the dirt out.". The idea closely aligned with the lesson idea: to improve water quality by adding something/treating it and filtering. In comparison, in a high school chemistry lesson that targeted ideas about lightning

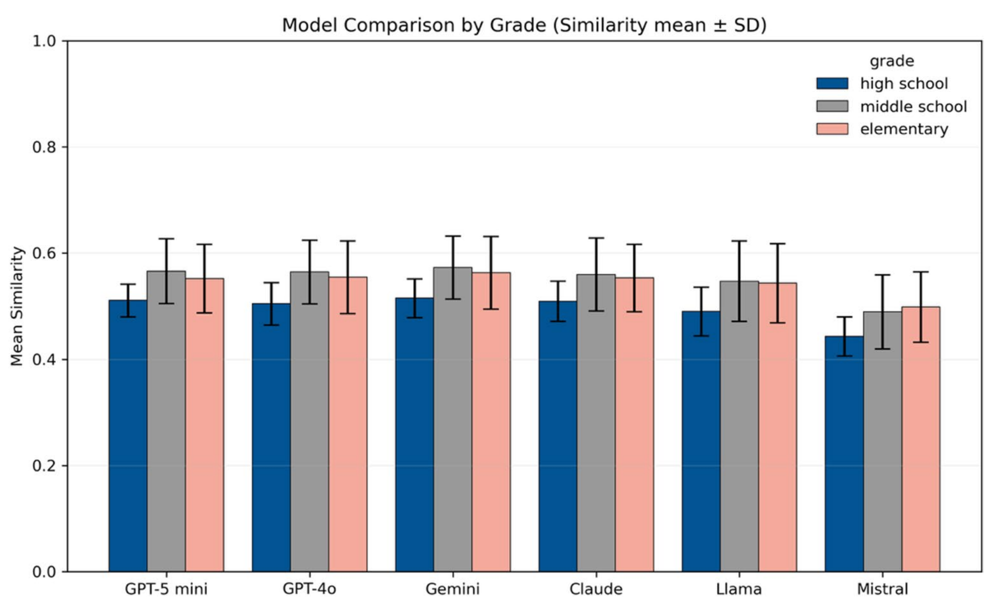


Fig. 6 Semantic similarity between simulated and lesson ideas across six models by grade

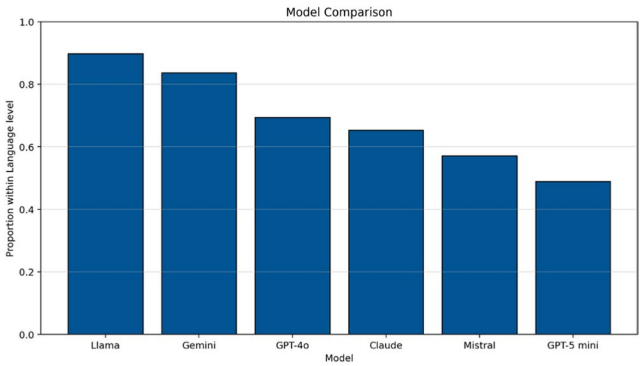


Fig. 7 Proportion of simulated ideas within language level across six LLMs

movement, the Gemini-generated ideas (e.g., "Our model had clouds, wind, and like, electricity stuff.") did not resemble the lesson ideas.

Note that we reported the raw cosine similarity values in this analysis, which represented comparisons at the idea-cluster level. Another approach is to determine a cutoff threshold for whether the LLM-simulated ideas aligned with lesson ideas (binary, yes/no). This idea-level analysis (see Appendix E) revealed similar patterns.

RQ1b: Language Level

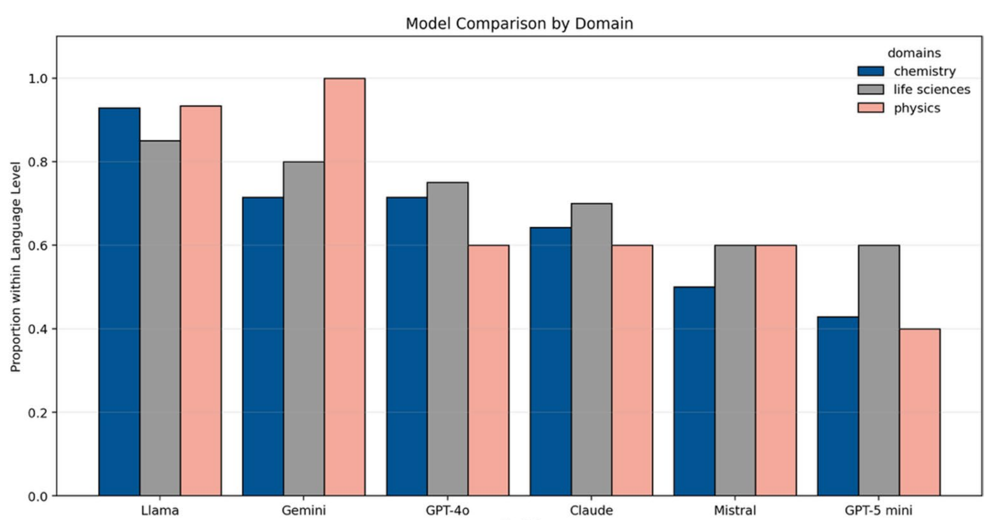
Next, we analyzed the language levels of LLM-generated ideas (Fig. 7). Overall, 69% of the simulated ideas were at or below the target students' grade level. A likelihood ratio test suggested that the model with LLM produced better fit than a model without ($\chi^2(5) = 57.44$, $p < .001$). Llama-4 performed best, with over 80% of its outputs meeting or falling below the expected grade level, which was statistically higher than Mistral-7B (Δ proportion = 0.33, OR = 67.57, 95% CI [10.33, 443], $z = 4.39$, $p < .001$) and GPT-5-mini (Δ proportion = 0.41; OR = 153, 95% CI [21.2, 1103], $z = 4.99$, $p < .001$).

The proportion of ideas with the appropriate language level did not differ significantly by domain (physics: 0.69, life sciences: 0.72, chemistry: 0.66; $\chi^2(2) = 0.51$, $p = .77$; Fig. 8). Within each model, the proportion of ideas within language level also showed no significant variation across domains.

Regarding grade levels (Fig. 9), a likelihood ratio test showed that the proportion of ideas within the appropriate language level significantly varied across grades, $\chi^2(2) = 58.54$, $p < .001$. Post-hoc analyses revealed significant pairwise differences: the proportion for high school and middle school was significantly higher than elementary school (high school: OR = 160, 95% CI [31.1, 819], $z = 6.08$, $p < .001$; middle school: OR = 3.16, 95% CI [1.29, 7.74], $z = 2.51$, $p = .01$). Within each model, Chi-square test indicated significant relations between the proportion of ideas within the appropriate language level and grade, Llama-4 ($\chi^2 = 6.91$, $p = .03$), Gemini-2.5-Flash ($\chi^2 = 14.92$, $p = .001$), GPT-4o ($\chi^2 = 18.82$, $p < .001$), Claude-Sonnet-4 ($\chi^2 = 21.79$, $p < .001$), Mistral-7B ($\chi^2 = 18.62$, $p < .001$), and GPT-5-mini ($\chi^2 = 38.49$, $p < .001$). These results indicated that LLM-generated ideas showed more complex language than what would be expected for lower grades.

A possible approach to improving these results is through iterative prompting. For responses that exceeded readability thresholds, we ran a follow-up prompt for the LLMs to "Revise this response to match the language expected for {grade_level}." Overall, most of the regenerated responses improved and fell at or below the target grade levels, Gemini-2.5-Flash (95.92%), Llama-4 (91.84%), Mistral-7B (91.84%), GPT-4o (89.80%), Claude-Sonnet-4 (83.67%), and GPT-5-mini (55.10%). Readability alignment was higher for high school grades (all LMMs reaching 100%) and showed more variability for elementary grades (range

Fig. 8 Proportion of simulated ideas within language level across Six LLMs by domain



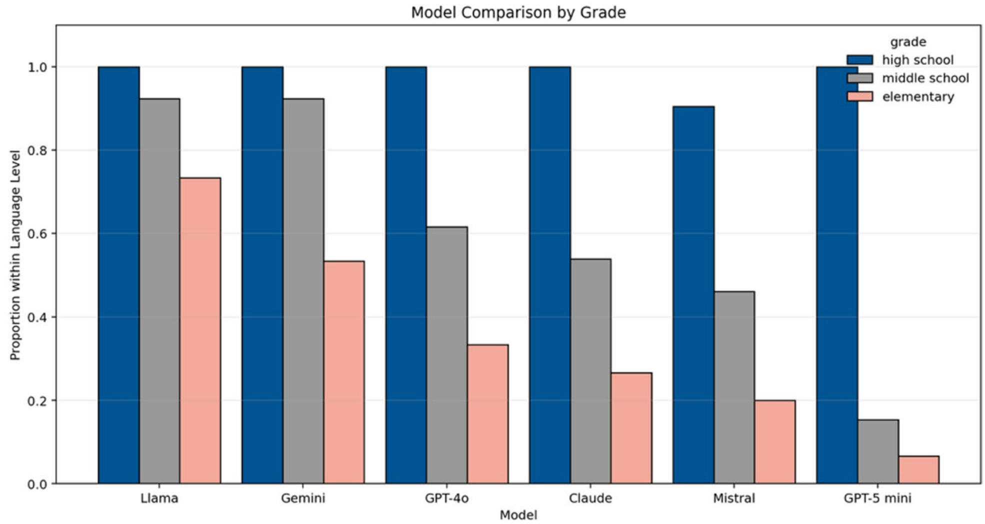


Fig. 9 Proportion of simulated ideas within language level across Six LLMs by grade

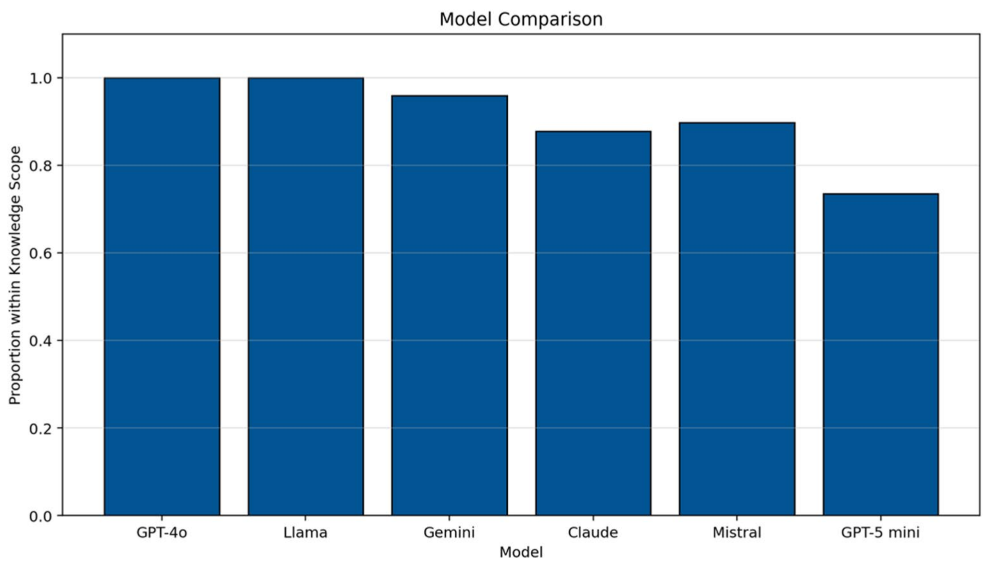


Fig. 10 Proportion of simulated ideas within knowledge scope across six LLMs

6.67%–80%, with low alignment for GPT-5-mini: 6.67% and Claude-Sonnet-4: 53.33%).

RQ1c: Knowledge Scope

Finally, we examined *knowledge scope*, or the proportion of simulated ideas that fell within the grade-level knowledge expectations, based on the lessons' learning objectives and prior instruction. Overall, 91% of the simulated ideas were within the corresponding knowledge scope (Fig. 10). Specifically, GPT-4o and Llama-4 performed the best: their simulated ideas fell within the expected knowledge scope for the target grades. In contrast, the ideas generated by GPT-5-mini were more likely to exceed the appropriate grade-level scope, compared to GPT-4o (Δ proportion = −0.27), Llama-4 (Δ proportion = −0.27), and Gemini-2.5-Flash (Δ proportion = −0.22). The proportion of simulated ideas within knowledge scope did not differ significantly by domain overall, likelihood ratio test: $\chi^2 = 0.06$, $p = .97$ (Fig. 11), and for each LLM.

Regarding grade levels (Fig. 12), a likelihood ratio test suggested that adding grade as a fixed effect improved the model's fit, $\chi^2 = 17.18$, $p < .001$. Compared to high school, the within-scope proportion was significantly lower for middle school (Δ proportion = −0.09; OR = 0.14, 95% CI [0.03, 0.68], $p = .02$) and elementary school (Δ proportion = −0.16; $OR = 0.07$, 95% CI [0.02, 0.33], $p < .001$). Chi-square tests for each LLM revealed significant grade-level differences only for GPT-5-mini ($\chi^2 = 9.18$, $p = .01$) and not the other LLMs. For example, in a fourth-grade lesson on energy transfer, GPT-5-mini generated ideas that might be overly complex, e.g., "If we zoomed in on a marble, you'd see atoms move but we couldn't see shape change".

Fig. 11 Proportion of simulated ideas within knowledge scope across Six LLMs by domain

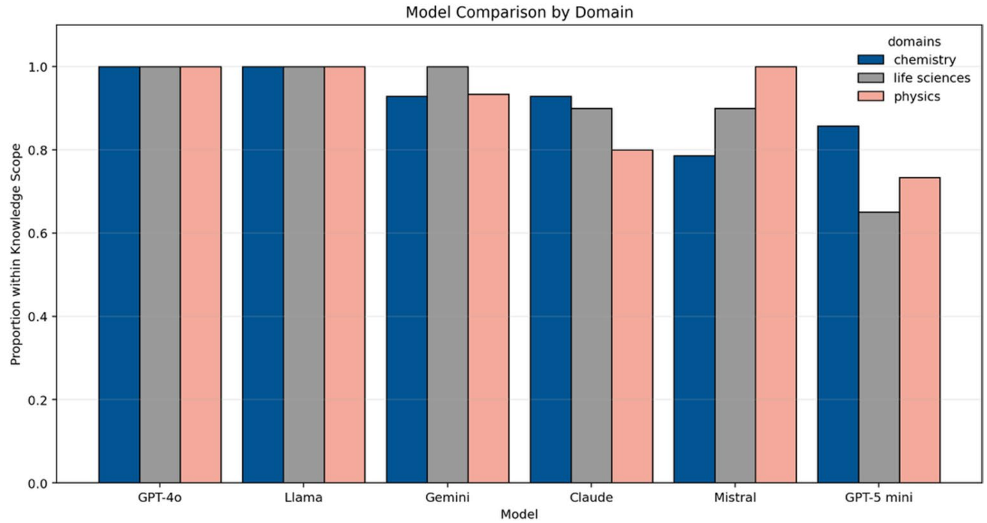


Fig. 12 Proportion of simulated ideas within knowledge scope across six LLMs by grade

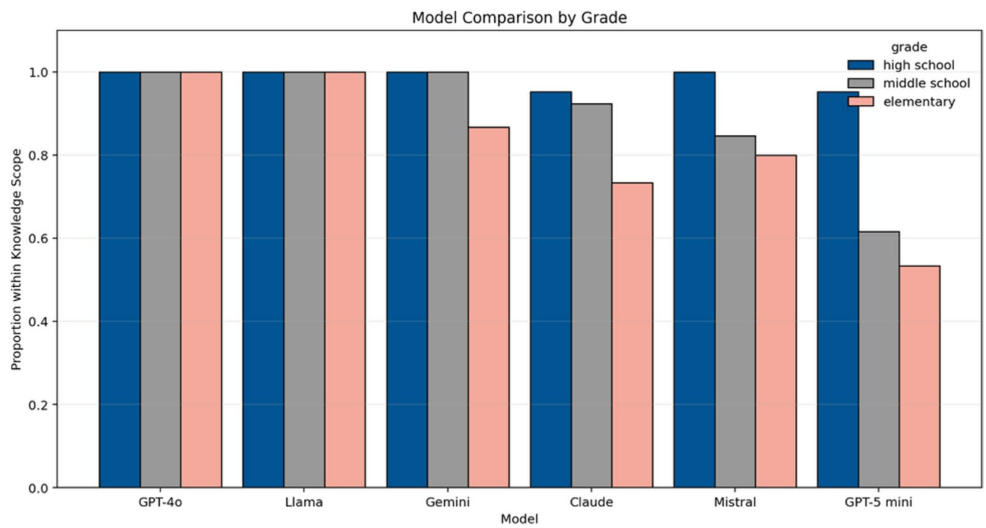


Table 2 Comparing lesson ideas and LLM-generated ideas

Theme	All LLMs	GPT-5-mini	GPT-4o	Claude 4	Gemini 2.5	Mistral-7B	Llama-4
Broad reasoning	60.54%	53.06%	71.43%	51.02%	65.31%	75.51%	57.14%
Misconception substance	17.69%	16.33%	20.41%	18.37%	14.29%	22.45%	14.29%
Technical vocabulary	35.71%	95.92%	30.61%	34.69%	10.20%	14.29%	28.57%
Less uncertainty	14.63%	16.33%	6.12%	22.45%	6.12%	12.24%	24.49%
Analogy use	7.48%	2.04%	8.16%	14.29%	6.12%	6.12%	8.16%
n	294	49	49	49	49	49	49

Idea Divergence

We qualitatively compared LLM-generated and lesson ideas to identify areas of divergence. Although the generated ideas generally overlapped with the targeted concepts in the lesson ideas, we observed that LLM ideas showed *less specific reasoning* grounded in lesson contexts, evidence, and mechanisms (60.54%; see Table 2 for range across LLMs). For instance, in a middle school lesson about ecosystems, the lesson ideas included specific examples of infrastructural and social factors linked to students' locations (e.g., dams, fire suppression, gardening), while the LLMs (e.g., Mistral) were more generic (e.g., "careless human activities").

Another area of divergence across LLMs is the substance of misconceptions in LLM-generated responses. Our prompts to the LLMs instructed them to include "partial or incorrect scientific reasoning". We noted that the misconceptions in LLM-generated ideas were *less developmentally appropriate* than the lesson ideas (17.69%). For example, in an elementary lesson about weather and hazards, the lesson ideas suggested conflicting claims about how much rain different regions got, which in turn limited the amount of fruit trees they could grow. These ideas might reflect realistic limitations in students' regional knowledge. In comparison, the LLM simulations (e.g., Gemini-2.5-Flash) produced naïve ideas or fundamental errors, e.g., "bananas are yellow, and yellow is a sunny color, so they need sun".

Finally, we noted several *differences in language patterns* between LLM-generated ideas and lesson ideas. First, some LLMs (e.g., GPT-4o, GPT-5-mini) produced longer sentences with more technical vocabulary than the lesson ideas (35.71%). For example, in a high school lesson about zombie fire, GPT-5-mini generated ideas such as "Behind the scenes microbes might help decompose peat and create gases that help smolder", whereas the lesson ideas were more informal, e.g., "There must be something about the peat that makes it able to burn under the ice". As another example, LLM-generated ideas included explicit notations uncommon in lesson ideas, e.g., "Photosynthesis equation is $6CO_2 + 6H_2O + \text{light energy} \rightarrow C_6H_{12}O_6 + 6O_2$". Second, the lesson ideas more frequently showed markers of uncertainty when students were developing hypotheses (e.g., "it seems like", "maybe"; 14.63%) than LLM-generated ideas. Finally, LLMs frequently output analogies and anecdotal ideas (7.48%), e.g., "Elephants suck up water and spray it, but that doesn't filter it". These analogies are less common in the lesson ideas. Together, findings suggest that while the generated ideas showed some overlap with the lesson ideas, they needed additional instruction to align with lesson contexts, students' reasoning, and language.

RQ2: Teachers' Perceptions of LLM-Generated Ideas and Instruction Suggestions

All teacher interviewees evaluated the LLM-generated responses based on authenticity (whether the responses reflected real student thinking) and usefulness (whether the responses informed teachers' understanding of student ideas). We briefly reported these insights to complement the quantitative analyses (RQ1).

Teachers stated that the responses were overall realistic and aligned with their lessons' objectives. They mentioned that the responses were most useful when they sparked teachers' sensemaking about what students knew and did not know. Rylie (pseudonym), who was drafting an eighth-grade physics lesson, showed sensemaking when analyzing the LLM-generated ideas about energy transfer at different points on a roller coaster track. In analyzing an LLM-created idea, "I think the coaster uses up all its energy at the bottom and then it's gone", she noted: "This is incorrect and is important for me to think about, if students think that this is the motion of the roller coaster and visualizing this motion." In response to another LLM-generated idea, "The coaster has both potential and kinetic energy in the middle since it's still a little high but also moving", she reflected:

> The fourth point is also helpful because it's showing the presence of both energy; it's so rare in the work that I've analyzed. Maybe one or two students have this idea that maybe two energy can be there at the same time. This idea shows that the student is reasoning too (it's still high but moving). I wish more of my students can get to this point.

Teacher Allan (eighth grade, physics) also showed deep sensemaking around LLM-generated responses to plan how he might explain the concepts to his students. He reasoned:

> I might want to have a conversation about what is gone. Is it the energy that is gone? If it's gone, how can it come back up? We want them to understand that it isn't gone, it just changes form.

Additionally, all teachers found value in the LLM-generated instructional prompts to scaffold student ideas and surface unexpected insights. Rylie (eighth grade, physics) appreciated suggestions for organizing student thoughts and giving students "experience anchors", such as asking for evidence, reframing questions, and providing counterexamples. Similarly, John (sixth grade, physics) noted unexpected suggestions that he had not considered in a modeling lesson about bird movement. "I wasn't thinking to include examples of penguins. […] It's nice to see how students might revoice their thinking and have these wonderings." Three teachers noted that these suggestions were valuable in preparing new lessons, as they could save time and help teachers anticipate possible student responses.

Teachers also identified areas of improvement for the generated ideas and instructional prompts. Two observed that the LLM appeared to use more advanced or unrealistic concepts and vocabulary. For example, in simulating ideas for a middle-school eighth-grade lesson on thermal energy, the LLM-generated ideas included phrases such as "loses cold quicker". John pointed out that these phrases might be unnatural for his students. These insights overlapped with RQ1 findings that the language in LLM-generated ideas did not always align with the target grade levels. Further, Sam (high school; AP environmental science) and Alyssa (elementary) mentioned that some LLM-generated prompts were overly leading. Alyssa noted: "The follow-up questions redirect the conversation based on the AI's chosen topics. It might be more meaningful if students had generated the topics." Finally, Jan (high school) highlighted the need for the questions to "ask how/why, not yes/no questions" to better elicit student thinking.

Teachers mentioned the need for refinement of LLM's outputs through iterative prompting with the LLMs to add lesson contexts, knowledge of their students (e.g., current level of conceptual understanding, prior assessment, discussion participation norms), and instructional goals. Sam noted: "It is rare that I can use AI-generated text from scratch. It needs to pick up context […] I might ask, can you write a question that gets at this, and then tweak it." Such refinement indicates the importance of teachers' oversight and idea co-development, particularly in aligning LLM-generated ideas with the targeted lessons' knowledge scope. As teachers gained clarity and ideas about student thinking (through LLMs-simulated ideas), they might refine the learning objectives and re-prompt the LLMs, leading to new ideas for learning facilitation.

Discussion

Responsive teaching frameworks (Hammer et al., 2012; Windschitl et al., 2012) guided how we designed the LLM prompt to generate different student ideas and varied levels of understanding and evaluated the ideas' value for instructional practice. We discuss implications for research and practice, specifically how to improve the quality of LLM simulations and support responsive teaching.

Implications for Research

Findings echo prior research that has demonstrated that LLMs can simulate students' ideas to some extent (Jin et al. 2025; MacNeil et al. 2024; Zheng et al. 2025a, b) but might be unsuccessful in representing baseline knowledge (Bhowmik et al., 2024) or language complexity (Zheng et al. 2025a, b). For RQ1, we find that the LLM-generated ideas exceed the readability and knowledge scope expected for elementary school and middle school students. Other researchers (Jin et al., 2025; Liu et al., 2023) also reported that LLMs achieved better performance in simulating advanced or high-knowledge students but struggled to replicate lower-knowledge students' misconceptions in mathematical reasoning. A possible explanation is that LLMs are not yet able to represent the reasoning process of younger students (Cherian et al., 2024). Our results indicate improved performance from a multi-prompt LLM pipeline, where we added additional steps to first evaluate the responses for alignment and then revise responses for specific grade levels. Future work can involve fine-tuning based on students' responses, including targeted fine-tuning on elementary students' discourse, additional prompting approaches such as chain-of-thought for LLMs to explain their reasoning step-by-step, or reinforcement learning and preference optimization to incorporate data annotations of desirable versus undesirable responses (Binz et al., 2025; Scarlatos et al., 2025). These approaches have shown promise for reflecting children's language and reasoning (Milička et al., 2024). In addition, while we only interviewed one elementary school teacher, researchers can involve more teachers in these grades to critique the responses and inform LLM prompting and training.

We propose a comprehensive evaluation combining similarity to lesson ideas, language level, knowledge scope, idea divergence and overlap, and practitioners' insights. Such evaluation is critical to make sure the LLMs represent student thinking accurately while supporting teacher sensemaking and instructional planning. For similarity, the average cosine similarity value comparing LLM-generated and lesson ideas is 0.52, $SD=0.07$. This value falls within the range of acceptable thresholds for idea similarity from prior studies, 0.42–0.70 (Cann et al., 2025; Penumatsa et al., 2006), indicating their potential utility in practice. Analyses of language levels and knowledge scopes offer additional insights to guide LLM selection. They reveal variations in performance across LLMs: Llama-4, Gemini-2.5-Flash, and GPT-4o are the most balanced, while Mistral shows the weakest performance overall. Although GPT-5-mini performs well in resembling lesson ideas, the model tends to exceed the expected language level and knowledge scope. These results provide references for researchers and practitioners to choose a suitable LLM for simulating student ideas across evaluation criteria. They also demonstrate the capacity of open-source models (e.g., Llama-4) that can be used in local, lower-resource applications.

Finally, we provide additional qualitative and teachers' insights to complement prior work that has used computational approaches to evaluate simulation performance (Xu et al., 2025). For our use case, teacher interviews (RQ2) reveal that diverging ideas can still be useful. Such ideas can prompt teachers to identify unexpected ideas, articulate assumptions about student understanding, and create plans to promote student thinking, even when teachers critique the authenticity of the simulated responses (Zheng et al. 2025a, b). Future studies can examine the models' instructional utility as an additional performance metric. For example, researchers can examine the numbers of teachers' revisions to lesson plans or changes to classroom facilitation, following teachers' interactions with the simulated ideas.

Implications for Practice

Findings from teacher interviews (RQ2) add nuances to the results about LLMs' promise and limitations (RQ1). They highlight considerations about (1) capturing authentic students' understanding that reflects their language and knowledge, and (2) revising LLMs' suggested follow-up moves for teachers to elicit student thinking, rather than leading students' responses. Results underscore the potential of technologies that simulate student ideas to support responsive teaching. Researchers have attempted to promote responsive teaching through collaborative video-based reflection (Abdulhamid, 2022; Ebby et al., 2023) and practice-based professional development (Kavanagh et al., 2020). These approaches have strong grounding in learning artifacts, but teachers may have limited time and resources to engage in repeated practice. We explore how to leverage emerging LLMs to promote responsive science teaching. Through encountering and interpreting possible student thinking in the LLM-simulated ideas, teachers can anticipate student thinking, reflect on teaching strategies, and adapt instruction in real-time interactions (Hu et al., 2025). They might engage in sensemaking to notice the disciplinary connections within students' ideas and plan how to respond to these ideas (Robertson & Richards, 2017).

While the LLM-simulated ideas provide a window into possible student thinking and classroom moves, teachers can improvise in instruction to accommodate or extend students' emergent sensemaking (Cherbow et al., 2025). In addition, in place of static LLM output, insights from teacher interviews highlight the potential for iterative prompting between teachers and the LLMs. Through these interactions, teachers can correct the output and clarify classroom contexts, including prior instruction, learning objectives, discussion norms, and students' existing knowledge. Increased experimentation with GenAI can shift teachers' roles from observers who have not actively engaged with the tools to collaborators and innovators who participate in the creation, evaluation, and refinement of GenAI-enabled learning activities (Zhai, 2025).

Our approach can be extended to pre-service teacher education, to help novice teachers practice recognizing and responding to varied student thinking (Pan et al., 2025). Other applications might include practice with dynamic, multi-turn dialogues with simulated students to approximate the complexity of real-time classroom discourse; and real-time support that makes salient student thinking and suggests follow-up instructional moves for teachers. To facilitate this integration, technology development should be paired with professional development opportunities for teachers. These opportunities can introduce teachers to responsive teaching vision, invite them to explore AI output, and offer guidance on how to critique and use AI simulations to inform instruction. We note the importance of institutional support to promote implementation, including the establishment of clear data privacy policies, such as what student information to include as input for LLMs. These efforts align with growing global interest in pedagogically meaningful and ethical approaches to integrating GenAI into science teacher education (Lee et al., 2026).

Limitations & Future Research

We acknowledge several limitations regarding study design and technical considerations. First, we provide exploratory insights into LLMs' capacity to simulate student ideas and have not examined the impact of the simulation on responsive teaching practices in real classrooms. Second, we acknowledge potential harms if teachers overly rely on inaccurate simulations, which might constrain rather than support responsiveness. Third, a construct validity consideration is that we compared the LLM-generated ideas to ideas from pilot data collection and teacher predictions, which might not fully represent student ideas.

These limitations highlight the need for future work that examines classroom implementation. Researchers can compare LLM-generated ideas with student ideas in authentic instructional contexts. They can also explore teachers' sensemaking and uptake, through observational studies of teacher planning and enactment, experimental designs to examine whether and how simulated ideas improve the quality of responsive teaching in science education, and longitudinal studies to investigate teachers' perceptions over time.

For technical considerations, we rely on zero-shot prompting to generate student ideas, without comparing different prompting strategies or fine-tuning methods. Future research can experiment with more advanced approaches, including finetuning with target students' discourse data and incorporating teachers' insights into model training, to enhance performance and educational applicability. For the knowledge scope analysis, we acknowledge potential systematic biases when an LLM was judging output from other LLMs and note the importance of robustness check (e.g., evaluating several LLMs) and human oversight. Further, beyond cognitive modeling, researchers can examine whether and how LLMs can simulate motivational, affective, and social aspects of student engagement in classroom interactions.

Conclusion

We explore how LLMs can simulate students' ideas in terms of semantic overlap with lesson ideas, knowledge scope, language level, and teachers' perceptions. We find that LLMs demonstrate moderate simulation performance, although performance varies by models and grade levels. LLMs tend to fall short in simulating the knowledge scope and language level of lower-grade students, as they might employ broader reasoning or more advanced concepts than expected. Despite these limitations, teachers generally hold positive perceptions, noting that simulated ideas can help them notice and build on students' ideas. Overall, our results highlight the potential benefits of using LLMs to simulate students' ideas to promote responsive teaching. We call for further research that investigates ways to enhance the accuracy of the simulated ideas and documents their impact on responsive teaching practices.

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Data Availability Data are available from the authors upon reasonable request.

Declarations

Ethical Approval This study was performed in line with the ethical standards of the Institutional Review Board (IRB #24-1769).

Informed Consent This study was performed in line with the ethical standards of the Institutional Review Board (IRB #24-1769).

Consent to Publish All participants have consented to publish their anonymized data.

Regarding Research Involving Human Participants The research involves human participants.

Competing interests The authors have no propriety or competing interests in any material discussed in this article.

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